
Overlapping Schwarz preconditioning: ASM (unscaled) vs. sASM (weighted)

A systematic study over dimension, boundary conditions and anisotropy on unstructured P1 FEM

Case a

full Dirichlet,
isotropic

Case b

full Dirichlet,
anisotropic

Case c

Dirichlet+Neumann,
isotropic

Case d

Dirichlet+Neumann,
anisotropic

CG + PCASM + ICC(0) · PETSc 3.24 · 2026-06-18

Why does more overlap sometimes *increase* iterations?

- **Classical view:** more overlap $\Rightarrow$ smaller $\kappa \Rightarrow$ fewer iterations ($\kappa \leq C(1 + H/\delta)$).
- **Anomaly:** with *unscaled* BASIC ASM + inexact ICC local solves + CG, more overlap $\Rightarrow$ *more* iterations (seen in cardioid Sys2/Sys3).
- **Why rarely seen:** the default is RAS (non-symmetric, used with GMRES, built-in partition of unity $\Rightarrow$ never over-counts). Keeping CG symmetric forces BASIC — which steps into the trap.
- **This study:** compare ASM vs sASM across 4 cases on iterations, time, and interpretable quantities (κ , spectrum, $\hat{N}$, ω , Dirichlet-info propagation).

ASM (BASIC)

$$M^{-1} = \sum_i R_i^\top A_i^{-1} R_i$$

Overlap region added repeatedly $\Rightarrow$ over-count $\hat{N}$.

sASM (weighted)

$$M^{-1} = D^{-\frac{1}{2}} \left(\sum_i R_i^\top A_i^{-1} R_i \right) D^{-\frac{1}{2}}$$

$D = \text{diag}(\text{multiplicity } m_k)$ removes $\hat{N}$, stays symmetric (CG-safe).

Method: equation, discretisation, subdomains, solver

Governing equation

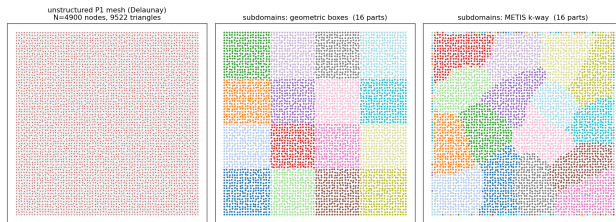
$$-\nabla \cdot (A \nabla u) = f \quad \text{on } \Omega = [0, 1]^d$$

A: diffusion tensor (iso/anisotropic)

Discretisation: P1 *unstructured* simplices (jittered Delaunay).

Solver & measurements

- Subdomains: S^d boxes *and* METIS k-way
- Local solve: ICC(0) (inexact) / Cholesky (exact)
- Krylov: CG (symmetric); overlap $O = 0 \dots 4$
- Measure: iters, setup/solve time, κ , spectrum, $\hat{N}$ (max multiplicity), $\omega = \max_i \lambda_{\max}(M_i^{-1} A_i)$



2D example: unstructured P1 mesh and the two subdomain partitions (boxes / METIS).

Definition of the four cases

Case	Coefficient A	Boundary condition	Dims
a	isotropic, $A = I$	full Dirichlet ($u = 0$ on all $\partial\Omega$)	1D/2D/3D
b	anisotropic $A = \text{diag}(\sigma_L, \sigma_T, \sigma_T)$ 2D: $\text{diag}(10, 1)$, 3D: Niederer $\sigma_L/\sigma_T = 7.58$ (fiber $\parallel x$)	full Dirichlet	2D/3D
c	isotropic, $A = I$	mixed: $x_0=0$ face Dirichlet, rest $\partial u/\partial n=0$	1D/2D/3D
d	anisotropic (as b)	mixed (as c)	2D/3D

Preconditioners compared:

ASM $M^{-1} = \sum_i R_i^T A_i^{-1} R_i$ vs.

sASM $M^{-1} = D^{-\frac{1}{2}}(\cdot) D^{-\frac{1}{2}}$

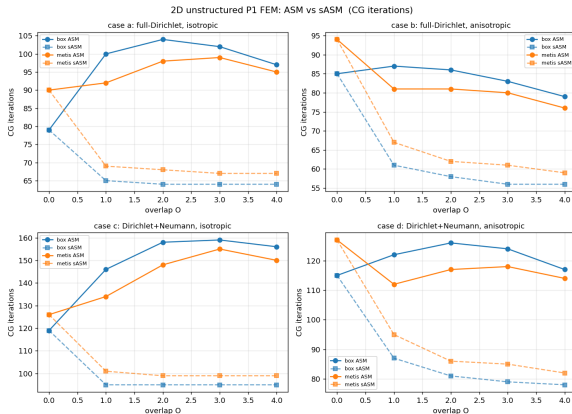
Comparison axes:

b vs a $\rightarrow$ effect of anisotropy

c vs a $\rightarrow$ effect of boundary condition

d vs c $\rightarrow$ anisotropy $\times$ mixed BC

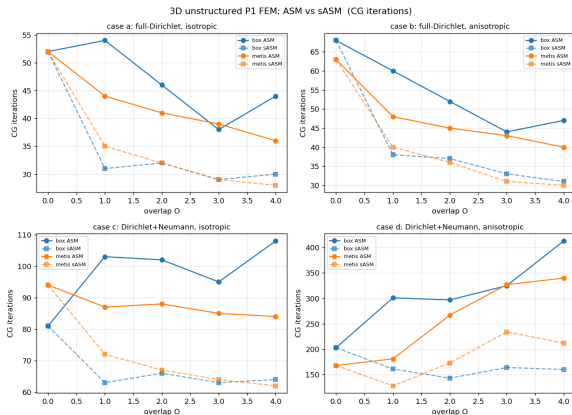
Overview: ASM vs sASM iteration count (2D)



- **What it shows:** per-case iteration–overlap curves, ASM vs sASM.
- ASM: overlap $\uparrow$ iters $\uparrow$ (anomaly) — a: 79 $\rightarrow$ 104, c: 119 $\rightarrow$ 158.
- sASM: monotone down and much lower — a: 79 $\rightarrow$ 64, c: 119 $\rightarrow$ 95.
- Mixed BC (c) anomaly stronger than full Dirichlet (a).
- Box and METIS partitions agree.

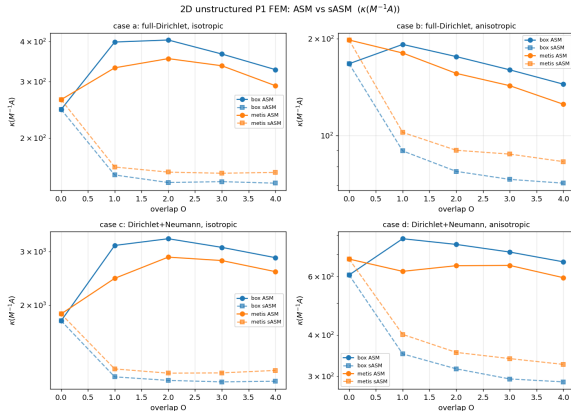
2D: CG iters vs overlap (solid = ASM, dashed = sASM; blue = boxes, orange = METIS).

ASM vs sASM (3D): the worst combination



3D ($N \approx 8000$, $\sim 49k$ tets): CG iters vs overlap. Solid = ASM, dashed = sASM.

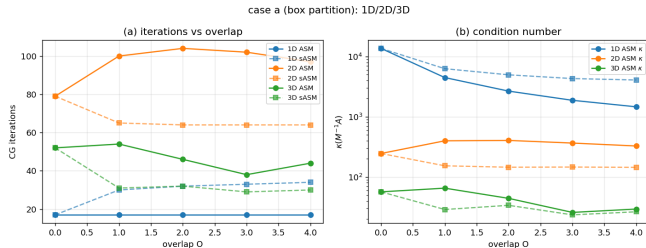
- **What it shows:**
iteration–overlap curves in 3D.
- Case d (mixed BC + anisotropy) worst: ASM
203 $\rightarrow$ 413 (doubles), sASM
203 $\rightarrow$ 143.
- sASM uniformly better in every 3D case.
- Geometric over-count $\hat{N}$ up to ~ 20 (box corners).
- ω rises to 2–5 on unstructured 3D P1 (weaker ICC).

Condition number $\kappa(M^{-1}A)$ (2D)

2D, per case: $\kappa(M^{-1}A)$ vs overlap (log axis). Solid = ASM, dashed = sASM.

- **What it shows:** preconditioned condition number vs overlap.
- ASM: κ rises / fails to drop ($\lambda_{\max}$ inflated by $\hat{N}$).
- sASM: clearly lower κ (over-count removed).
- Iteration counts track κ — mutually consistent.

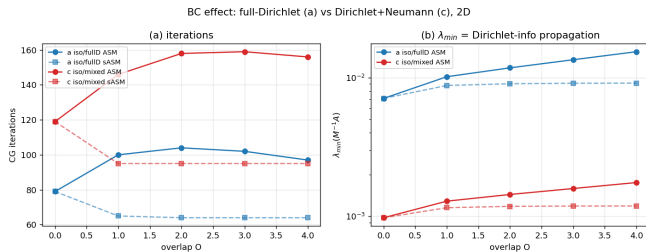
Dimensional effect: 1D immune, 2D/3D anomalous



Case a (full Dirichlet, isotropic, boxes): iters and κ for 1D/2D/3D.

- **What it shows:** fix case a, compare 1D/2D/3D.
- 1D: ICC(0) exact on tridiagonal
 $\Rightarrow \omega=1 \Rightarrow$ no anomaly (ASM iters flat at 17).
- In 1D sASM is worse (17 $\rightarrow$ 34): no over-count to remove, weighting only perturbs.
- 2D/3D: $\omega > 1$, over-count $\hat{N}=2^d$ acts
 $\Rightarrow$ anomaly; sASM fixes it.
- Dimensional amplifier is geometric
 $\hat{N}=m_{\text{axis}}^d$, not ω (ω saturates).

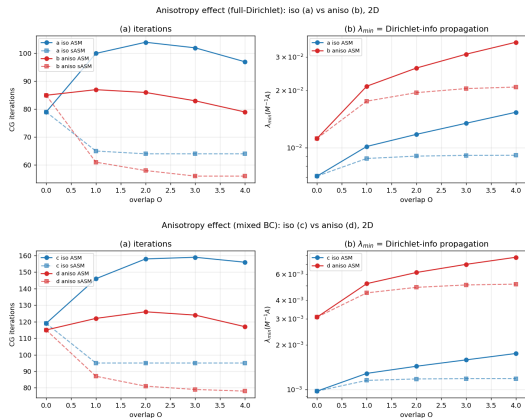
Boundary-condition effect: Dirichlet-info propagation



2D: full Dirichlet (a) vs mixed BC (c): iters and $\lambda_{\min}$ vs overlap.

- **What it shows:** how the BC changes the anomaly and $\lambda_{\min}$.
- Full Dirichlet (a): whole boundary pinned, info enters from all sides $\Rightarrow$ larger $\lambda_{\min}$, milder anomaly.
- Mixed (c): only one Dirichlet face; Neumann directions shrink $\lambda_{\min}$, slow propagation $\Rightarrow$ stronger anomaly.
- Overlap raises $\lambda_{\min}$ (faster info transfer) but ASM's $\hat{N}$ over-count cancels the gain.

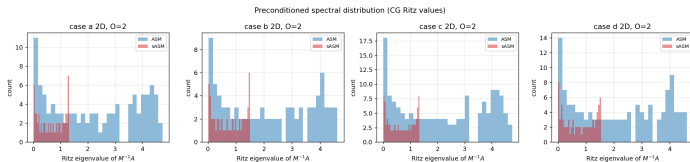
Anisotropy effect: raises ω and iterations, compounds with mixed BC



Top: full Dirichlet a (iso) vs b (aniso). Bottom: mixed BC c vs d.

- **What it shows:** effect of anisotropy on iters / $\lambda_{\min}$ under full Dirichlet and mixed BC.
- Anisotropy (2D $10\times$, 3D Niederer 7.58) raises the local ICC amplification ω and the iteration count.
- Geometric over-count $\hat{N}$ is coefficient-independent (pure geometry); anisotropy does not change it.
- Worst is case d-3D (anisotropy $\times$ mixed BC): ASM iterations double.
- sASM removes $\hat{N}$ and fixes the overlap *trend*, but its geometric weighting is coefficient-blind — under strong anisotropy / high contrast absolute κ stays high, calling for a spectral coarse space (GenEO).

Spectral distribution (CG Ritz values)

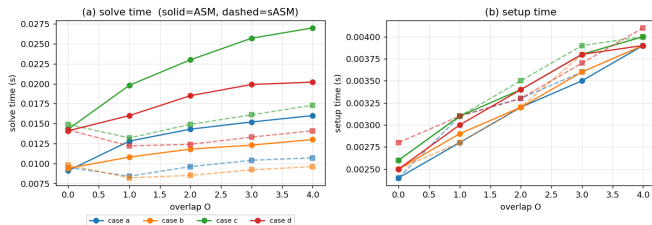


Ritz eigenvalues of $M^{-1}A$ at $O=2$. Blue = ASM, red = sASM.

- **What it shows:** the preconditioned spectrum.
- ASM: a cluster of large eigenvalues to the right ($\hat{N}$ pushes $\lambda_{\max}$ up).
- sASM: top end pulled back to $\approx \omega$, spectrum tighter $\Rightarrow$ faster CG.
- Spectral width = condition number = iteration gap.

Compute time: sASM slightly dearer per step, faster overall

2D (box): wall-clock time, ASM vs sASM



2D: solve and setup wall time vs overlap. Solid = ASM, dashed = sASM.

- **What it shows:** solve / setup time for ASM and sASM.
- sASM adds only two vector scalings ($D^{-1/2}$) per iteration — negligible overhead.
- Fewer iterations $\Rightarrow$ sASM total solve time usually shorter.
- Setup comparable (sASM reuses the same BASIC kernel + one multiplicity pass).
- Serial timing here (relative comparison); parallel trend on HPC is the same.

Conclusions

- 1 sASM beats ASM in 2D/3D across all cases:** removes the geometric over-count $\hat{N}$ — iterations down, κ lower, total time shorter, symmetry preserved (CG-safe).
- 2 1D is the exception:** ICC(0) exact ($\omega=1$), no over-count $\Rightarrow$ no anomaly — there sASM is slightly worse. sASM is a $d \geq 2$ fix.
- 3 Worst combination:** 3D + mixed BC + anisotropy (case d-3D), where ASM iterations *double* with overlap.
- 4 Boundary conditions:** full Dirichlet lets info enter from all sides $\Rightarrow$ larger $\lambda_{\min}$, milder anomaly; mixed BC the opposite.
- 5 Mechanism:** $\lambda_{\max}(\text{ASM}) \approx \omega \cdot \hat{N}$; $\hat{N}$ = geometric over-count (dimensional amplifier), ω = inexact-ICC factor (on/off switch).
- 6 Boundary of the fix:** sASM's geometric weighting is blind to anisotropy / high contrast — those need a spectral coarse space (GenEO).

Reproduction and tooling

Pipeline (PETSc 3.24)

```
python3 fem_build.py <a|b|c|d> <dim> <n> <nsub>
  unstructured P1 mesh (Delaunay) + assembly +
  both partitions → PETSc binary
```

```
mpicc schwarz_fem.c -lpetsc -o schwarz_fem
bash run_fem.sh # all cases × {box,METIS} × {ASM,sASM} × O
python3 parse_fem.py # → CSV + figures
```

Measured: iter, t_{setup} , t_{solve} , κ , Ritz spectrum, $\hat{N}$, ω .

Notes

- Serial + explicit S^d /METIS subdomains, independent of rank count.
- PETSc 3.24 API — runs as-is on HPC spack PETSc 3.24 at scale.
- High-contrast / anisotropic local ICC uses `positive_definite` shift.
- Next: add a GenEO spectral coarse space at the case-d boundary.